TAK Server Mission API Definition (Data Sync)

Overview

The Data Sync / Mission API provides the capability in TAK Server for clients, such as ATAK, to share collections of information which are tracked as part of missions. Once a mission has been created by a client, it immediately becomes available to other clients. There are three types of content that can be associated with a mission: CoT tracks, enterprise sync resources and mission logs. CoT tracks are referenced by their UID, so that any information that can be represented in a mission. Once CoT UIDs have been added to a mission, the server can then provide the client with the latest CoT event associated with each UID. Mission Logs are a series of free-form text entries that can be viewed individually or as a complete log.

Files can be uploaded to TAK Server through the Enterprise Sync API. The adds generality such that any type of file that is useful from a client perspective can be added to a mission. Common files types include GRGs, images, and KML (Keyhole markup language) files. These resources are referenced by calculating a deterministic hash for each file, based on the binary contents of the file. Once the association has been made between a mission and a file, the file contents of the mission can be searched by parameters including time interval and geographic location.

A substantial portion the API and related functions have been developed in order to provide clients with the ability to add and remove relevant mission content, determine what changes to missions have been done by other clients, and then request updates for only what each client requires. In this way, each client can always maintain a consistent local view of missions and their contents, which conserving resources such as local storage and network bandwidth.

In addition to CoT tracks and files, each mission can also contain a "Mission Log" (also referred to as a "Recce Log"). Mission logs are composed of individual log entries, which clients post to the server. Once a log entry has been added, it immediately becomes available as part of the mission log, in a similar fashion to enterprise sync resources and CoT tracks. CoT UIDs and file resources can also be associated with individual log entries, just as they can be associated with missions. Log entries may also be updated and deleted.

Similarly, tracks and files can also be removed from the mission. Every change to the state of a mission, including the creation or deletion of the mission itself, is tracked by TAK Server. Based on this information, the server can generate a change log detailing each change to the mission over time. This log can be filtered based on an arbitrary time interval.

Clients may subscribe to one or more missions. Once a client has subscribed to a mission, the server will send it a CoT message informing the client of each change. Based on this notification, the client can then request a change log for an appropriate time range.

Missions and related information are made available to clients via a REST (Representational State Transfer) API (Application Programming Interface). This approach models the mission information as a hierarchy of resources. For example, the "mission" resource contains the "content" sub-resource. API information is encoded as JSON (JavaScript? Object Notation) objects. The REST approach applies the standard HTTP (Hypertext Transfer Protocol) verbs GET, POST, PUT and DELETE to mission resources. These verbs correspond, respectively, to the operations "read", "create", "update" and "delete" (CRUD). Request and response

headers are also used appropriately, in accordance with the HTTP 1.1 standard. REST operations follow best practices with regard to safety and idempotency. For example, multiple concurrent requests to create a mission of the same name will not be afflicted by concurrency problems such as race conditions or deadlock. This hold true whether the multiple "create mission" requests come from the same client, or multiple clients.

API Response Format - Success

Successful API responses are encoded as JSON, and follow a standard format. This is intended to make parsing on the client-side (Web / Android) easier, no matter what JSON library or parsing method is used.

The response is a JSON object. The properties of the root object represent the following:

version: API version type: Type of data contained in the data array data: A JSON array of objects, of the indicated type. If the REST request result is empty, the array will be empty (but not null). In the case of a single response, there will be one element in the array. For a response containing multiple items, these items will be in the array

```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "pittsburgh",

      "keywords": [],

      "creatorUid": "bob",

      "createTime": "2016-05-03T15:22:44.107Z",

      "uids": [],

      "contents": []

    }

  ]

}
```

API Response Format - Error

Error responses are also encoded in a standard JSON response format. This enables a standard HTTP result handler to be used on the client-side when a error HTTP response code is received (such as 404 Not Found or 400 Bad Request).

The properties of the root response JSON object are:

status: A textual status code. code: a number error code. message: a human-readable message, often from the server-side exception.

```
{

  "status": "NOT_FOUND",

  "code": 1,

  "message": "Not Found: Mission named 'pittsburw' not found"

}
```

Response Headers

Except for binary file responses, API Responses are JSON documents, so unless noted otherwise, response headers will include:

```
Content-Type: application/json;charset=UTF-8
```

Timestamps

Timestamps are in the ISO-8601 format, with milliseconds.

Example: 2016-04-22T15:26:17.260Z

Mission API

Resource: Missions

The data sync mission API represents missions and their contents as REST resources (and subresources.)

```
api/missions
```

GET all missions

```
curl "http://<username>:<password>@localhost:8080/Marti/api/missions"
```

Success Response Code: 200 OK

```
{

  "version": "1.0.0",
```

```
  "type": "Mission",
  "data": [
    {
      "name": "train",
      "keywords": [],
      "creatorUid": "bob",
      "createTime": "2016-04-22T15:26:17.260Z",
      "uids": [
        {
          "data": "ANDROID-48:5A:3F:52:29:78",
          "timestamp": "2016-05-02T14:11:00.888Z",
          "creatorUid": "joe1234"
        },
        {
          "data": "ANDROID-00:AE:FA:7D:63:EB",
          "timestamp": "2016-05-02T14:11:00.896Z",
          "creatorUid": "joe1234"
        },
        {
          "data": "ANDROID-FC:C2:DE:4F:77:D6",
          "timestamp": "2016-05-02T14:11:00.903Z",
          "creatorUid": "joe1234"
        }
      ],
      "contents": [
        {
          "data": {
            "keywords": [],
```

```
            "name": "Wide_photo.kmz",
            "submissionTime": "2016-04-22T15:31:02.829Z",
            "submitter": "admin",
            "uid": "9698hhsg",
            "creatorUid": "yyy666",
            "hash":
"78cd540c8e352e6ff416fc1ec48de602c20a6274d1c1201409a5fbb107f23f8f",
            "size": 55981
          },
          "timestamp": "2016-04-22T15:31:47.800Z",
          "creatorUid": "joe1234"
        }
      ]
    },
    {
      "name": "pittsburgh",
      "keywords": [],
      "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D",
      "createTime": "2016-04-22T17:13:42.437Z",
      "uids": [
        {
          "data": "3b7b1005-f1de-43e6-bb23-7262365753a4",
          "timestamp": "2016-04-22T19:52:31.137Z",
          "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D"
        },
        {
          "data": "166f40da-0ba0-40dc-a297-f704e518ce20",
          "timestamp": "2016-04-22T17:19:28.19Z",
```

```
          "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D"
        },
        {
          "data": "f1774536-5e19-459b-865d-dba9c427d35f",
          "timestamp": "2016-04-22T20:10:02.401Z",
          "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D"
        }
      ],
      "contents": [
        {
          "data": {
            "keywords": [],
            "mimeType": "image/jpeg",
            "name": "20160421_145952.jpg",
            "submissionTime": "2016-04-22T19:52:30.572Z",
            "submitter": "uploader",
            "uid": "0d69da09-9ec5-49cb-bfe4-5c486f81beee",
            "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D",
            "hash":
"aba4714c08f03b41a990da90013fd0646f1f11fcaa62a2891bb23d2240dfd402",
            "size": 2620798
          },
          "timestamp": "2016-04-22T19:52:30.918Z",
          "creatorUid": ""
        },
        {
          "data": {
            "keywords": [],
```

```
            "mimeType": "application/xml",
            "name": "ActionBars.xml",
            "submissionTime": "2016-04-22T18:06:04.359Z",
            "submitter": "uploader",
            "uid": "e34ed860-a180-495b-af35-54249c1d76d1",
            "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D",
            "hash":
"6035329a18775e74be5d3e2a82b1934b3f279ea57c949d5cab694144e4dca3a3",
            "size": 80819
          },
          "timestamp": "2016-04-22T18:06:04.567Z",
          "creatorUid": ""
        },
        {
          "data": {
            "keywords": [],
            "mimeType": "text/plain",
            "name": "ATAKCRASH_2016_02_05_23_24_21.txt",
            "submissionTime": "2016-04-25T02:38:20.896Z",
            "submitter": "uploader",
            "uid": "1925fddf-6490-4076-8736-c05023ab6444",
            "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D",
            "hash":
"0badea48e7763750c469f676307dda5c8b01d6f67a599b6fa7395af5167d4c91",
            "size": 46758
          },
          "timestamp": "2016-04-25T02:38:21.123Z",
          "creatorUid": ""
        }
```

```
      ]

    }

  ]

}
```

GET (single mission)

```
curl
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh"
```

Success Response Code: 200 OK


```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "pittsburgh",

      "keywords": [],

      "creatorUid": "ANDROID-88:32:9B:AF:8D:56",

      "createTime": "2016-04-26T19:22:42.917Z",

      "uids": [

        {

          "data": "3b7b1005-f1de-43e6-bb23-7262365753a4",

          "timestamp": "2016-04-29T20:23:24.89Z",

          "creatorUid": ""

        },

        {

          "data": "76557dc7-5375-41ea-b645-43db91f35ee2",

          "timestamp": "2016-04-26T23:01:01.358Z",
```

```
        "creatorUid": "ANDROID-88:32:9B:AF:8D:56"
      },
      {
        "data": "e508393d-dce5-499c-af0e-11b030ba566e",
        "timestamp": "2016-04-26T19:38:17.827Z",
        "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D"
      },
      {
        "data": "744284b9-3ccc-40ac-983c-f33dd2f39f88",
        "timestamp": "2016-04-26T19:30:45.671Z",
        "creatorUid": "ANDROID-88:32:9B:AF:8D:56"
      },
      {
        "data": "9276f5f3-afd7-44c0-98d7-09aca7eae834",
        "timestamp": "2016-04-26T23:00:39.384Z",
        "creatorUid": "ANDROID-88:32:9B:AF:8D:56"
      },
      {
        "data": "f7702d71-3c61-40f3-86fc-8c2367b8473c",
        "timestamp": "2016-04-26T20:06:42.596Z",
        "creatorUid": "ANDROID-88:32:9B:AF:8D:56"
      }
    ],
    "contents": [
      {
        "data": {
          "keywords": [],
          "mimeType": "image/png",
```

```
            "name": "Screenshot_2015-02-03-14-23-54.png",

            "submissionTime": "2016-04-26T22:55:59.912Z",

            "submitter": "uploader",

            "uid": "1f8e0a9b-ea83-4ead-8f8b-d8f960b3df9f",

            "creatorUid": "ANDROID-88:32:9B:AF:8D:56",

            "hash":
"9311a6e2ae6383c4a892c185a542b7010128e31846a693513e1c9ad1794c9719",

            "size": 874598

          },

          "timestamp": "2016-04-29T18:22:46.109Z",

          "creatorUid": ""

        },

        {

          "data": {

            "keywords": [],

            "mimeType": "image/jpeg",

            "name": "20160413_175826.jpeg",

            "submissionTime": "2016-04-26T22:04:47.658Z",

            "submitter": "uploader",

            "uid": "2c632b65-97f2-4585-9837-e17430a59ec3",

            "creatorUid": "ANDROID-88:32:9B:AF:8D:56",

            "hash":
"d992fda4830ae1eb4864e3d96eb7c64b98ff72116cc51f5d168d9b22dd1f99c2",

            "size": 63536

          },

          "timestamp": "2016-04-26T22:04:47.878Z",

          "creatorUid": ""

        },

        {
```

```
          "data": {
            "keywords": [],
            "mimeType": "image/jpeg",
            "name": "20160426_200531.jpg",
            "submissionTime": "2016-04-26T20:06:52.217Z",
            "submitter": "uploader",
            "uid": "fd47701e-1ca9-40b1-abd0-0012f7c66769",
            "creatorUid": "ANDROID-88:32:9B:AF:8D:56",
            "hash":
"4bd35b9a075e7c30aae291042bf4fdefbbd892781111e4016d7d0352c163ee65",
            "size": 571678
          },
          "timestamp": "2016-04-26T20:06:52.464Z",
          "creatorUid": ""
        },
        {
          "data": {
            "keywords": [],
            "mimeType": "application/octet-stream",
            "name": "truststore-demo.p12",
            "submissionTime": "2016-04-26T23:01:47.815Z",
            "submitter": "uploader",
            "uid": "e551f55c-3b7a-46a3-a723-a23189212696",
            "creatorUid": "ANDROID-C0:BD:D1:5D:4A:3D",
            "hash":
"78210cee41bffac6b33b932f6af03d2b994399465e198052e6891015475afbca",
            "size": 1136
          },
          "timestamp": "2016-04-26T23:01:48.15Z",
```

```
          "creatorUid": ""

        }

      ]

    }

  ]

}
```


Error Response Code: 404 Not Found

Error Response Code: 400 Bad Request

GET mission CoT

```
curl
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/cot"
```

Fetch the latest CoT event for each content UID that is part of the mission.

Success Response Code: 200 OK

Response Body


```
<?xml version='1.0' encoding='UTF-8' standalone='yes'?>

<event version='2.0' uid='ANDROID-48:5A:3F:52:29:78' type='a-f-G-U'
time='2016-05-02T14:11:29Z' start='2016-05-02T14:11:29Z' stale='2016-
05-02T14:12:13Z' how='h-e'>

    <point lat='38.87925058057263' lon='-77.10781575234559'
hae='56.5731172456302' ce='9999999.0' le='9999999.0' />

    <detail>

        <contact phone="15712128188" endpoint="10.0.1.6:4242:tcp"
callsign="The Dude"/>

        <uid Droid="The Dude"/>

        <__group name="Rad Sensor" role="Team Lead"/>

        <status battery="59"/>

        <track speed="0.0" course="278.41118524401173"/>
```

```
        <precisionlocation geopointsrc="User" altsrc="DTED0"/>

        <_flow-tags_ demo1="2016-05-02T10:11:29Z"/>

    </detail>

</event>

<event version='2.0' uid='ANDROID-00:AE:FA:7D:63:EB' type='a-f-G-U-C-
I' time='2016-05-02T14:09:41Z' start='2016-05-02T14:09:41Z'
stale='2016-05-02T14:15:56Z' how='h-g-i-g-o'>

    <point lat='0.0' lon='0.0' hae='9999999' ce='9999999.0'
le='9999999.0' />

    <detail>

        <contact endpoint="192.168.105.200:4242:tcp" callsign="POLE
VAULTER"/>

        <uid Droid="POLE VAULTER"/>

        <__group name="Cyan" role="Team Member"/>

        <status battery="0"/>

        <track speed="0.0" course="28.459523203155857"/>

        <precisionlocation geopointsrc="???" altsrc="???"/>

        <_flow-tags_ demo1="2016-05-02T10:09:41Z"/>

    </detail>

</event>

<event version='2.0' uid='ANDROID-FC:C2:DE:4F:77:D6' type='a-f-G-U-C'
time='2016-05-02T14:08:19Z' start='2016-05-02T14:08:19Z' stale='2016-
05-02T14:14:34Z' how='m-g'>

    <point lat='35.1573591' lon='-79.0020032' hae='72.3514144121611'
ce='65.0' le='9999999.0' />

    <detail>

        <contact phone="+19108186543" endpoint="10.77.168.92:4242:tcp"
callsign="auman_dev_S5"/>

        <uid Droid="auman_dev_S5"/>

        <__group name="Cyan" role="Team Member"/>

        <status battery="0"/>

        <track speed="0.0" course="53.221692106904314"/>
```

```
        <precisionlocation geopointsrc="GPS" altsrc="GPS"/>

        <_flow-tags_ demo1="2016-05-02T10:08:19Z"/>

    </detail>

</event>
```

Error Response Code: 404 Not Found

Create Mission

Create a mission with the specified name.

Empty request body (all required information is contained in the URI.)

Query Parameters

optional

creatorUid : Client UID

tool : name of the tool this mission is associated with, i.e, “citrap”

Success Response Code: 201 Created

Example: Create a mission named "Pittsburgh" and a creatorUid of "Bob"

```
curl -X PUT
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh?creatorUid=bob"
```

Response Body:

```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "pittsburgh",

      "keywords": [],

      "creatorUid": "bob",
```

```
      "createTime": "2016-05-03T15:22:44.107Z",

      "uids": [],

      "contents": []

    }

  ]

}
```

The create action is idempotent: if a mission with the specified name already exists, the call will still succeed.

DELETE (delete mission named 'xyz')

```
curl -X DELETE
"http://<username>:<password>@localhost:8080/Marti/api/missions/xyz"
```

Success Response Code: 200 OK

```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "xyz",

      "createTime": "2016-01-12T20:42:45.755Z",

      "contents": [],

      "keywords": []

    }

  ]

}
```

Error Response: 404 Not Found

```
{
```

```
  "status": "NOT_FOUND",

  "code": 1,

  "message": "Not Found: Mission named 'xyz' does not exist"

}
```

Resource: Mission Content

PUT (add mission content with hashes "12fg" and "789n" to mission pittsburgh)

Enterprise Sync resources (files) can be referenced by hash and / or uid.

by hashes:

```
curl -X PUT
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/contents" -H "Content-Type: application/json" -d '{"hashes" :
["12fg", "789n"]}'
```

by uids:

```
curl -X PUT
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/contents" -H "Content-Type: application/json" -d '{"uids" :
["asdf", "zxv"]}'
```

both:

```
curl -X PUT
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/contents" -H "Content-Type: application/json" -d '{"hashes" :
["12fg", "789n"], "uids" : ["asdf", "zxv"]}'
```

Required Request Headers:

```
Content-Type: application/json
```

JSON Request Body:

```
{

  "hashes" : [ "<hash>", "<hash>" ],

  "uids" : ["<uid>", "<uid>" ]
```

```
}
```

Request Parameters *Optional*

creatorUid : Client UID

Success Response Code: 200 OK


```
{
 "version": "1.0.0",
 "type": "MissionChange",
 "data": [
   {
     "type": "CREATE_MISSION",
     "missionName": "hamburger",
     "timestamp": "2016-04-19T19:58:21.526Z",
     "creatorUid": "joe1234"
   },
   {
     "type": "ADD_CONTENT",
     "contentResource": {
       "keywords": [],
       "mimeType": "application/x-zip-compressed",
       "name": "gpx_test.zip",
       "submissionTime": "2016-02-18T16:39:31.282Z",
       "submitter": "uploader",
       "uid":
"004ac8ddcfe51028676ae81e0cdbf39c62dfeca6a9e55a7de2c92dc379cf6759",
       "hash":
"004ac8ddcfe51028676ae81e0cdbf39c62dfeca6a9e55a7de2c92dc379cf6759",
       "size": 3021
     },
```

```
    "missionName": "hamburger",
    "timestamp": "2016-04-19T20:01:13.59Z",
    "creatorUid": "joe1234"
  },
  {
    "type": "ADD_CONTENT",
    "contentUid": "abc1234",
    "missionName": "hamburger",
    "timestamp": "2016-04-19T20:09:45.151Z",
    "creatorUid": "joe1234"
  }
]
```

DELETE (remove mission content by a single hash or uid)

Query Parameters

hash : Hash of file to remove.

uid : UID to remove

(either hash or uid must be specified, not both)

Optional

creatorUid : Client UID

```
curl -v -X DELETE
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/contents?hash=789n"

curl -v -X DELETE
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/contents?uid=xyz"
```

Success Response Code: 200 OK

Response Body:

JSON representation of the contents of the mission, after the removal


```
{
  "version": "1.0.0",
  "type": "Mission",
  "data": [
    {
      "name": "pittsburgh",
      "createTime": "2016-01-22T22:27:26.288Z",
      "contents": [
        {
          "filename": "",
          "keywords": [],
          "mimeType": "image\/png",
          "name": "",
          "submissionTime": "2016-01-22T22:27:57.908Z",
          "submitter": "",
          "uid": "e4da5c6f9137463d99f0764e0c7b1011",
          "hash": "12fg",
          "size": 663707860
        }
      ],
      "keywords": []
    }
  ]
```


Error Response Code: 404 Not Found (if the mission doesn't exist)

Resource: Mission Changes

GET (get changeset for mission pittsburgh)

```
curl
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/changes"
```

Query Parameters:

secago : time interval - number of seconds in the the past, until now.
start : start of time interval. Takes precedence over "secago" if both parameters are specified.
end : end of time interval

Success Response Code: 200 OK


```
{

  "version": "1.0.0",

  "type": "MissionChange",

  "data": [

    {

      "type": "ADD_CONTENT",

      "contentUid": "ANDROID-FC:C2:DE:4F:77:D6",

      "missionName": "train",

      "timestamp": "2016-05-02T14:11:00.903Z",

      "creatorUid": "joe1234"

    },

    {

      "type": "ADD_CONTENT",

      "contentUid": "ANDROID-00:AE:FA:7D:63:EB",

      "missionName": "train",

      "timestamp": "2016-05-02T14:11:00.896Z",

      "creatorUid": "joe1234"

    },

    {
```

```
      "type": "ADD_CONTENT",

      "contentUid": "ANDROID-48:5A:3F:52:29:78",

      "missionName": "train",

      "timestamp": "2016-05-02T14:11:00.888Z",

      "creatorUid": "joe1234"

    },

    {

      "type": "REMOVE_CONTENT",

      "contentUid": "ANDROID-10:A5:D0:FC:24:9B",

      "missionName": "train",

      "timestamp": "2016-05-02T14:09:19.408Z",

      "creatorUid": "bob"

    },

    {

      "type": "REMOVE_CONTENT",

      "contentUid": "WinTec.73ed908c-fc60-406f-8260-
8f060e667108.NMEA.SA",

      "missionName": "train",

      "timestamp": "2016-05-02T14:09:02.333Z",

      "creatorUid": "bob"

    },

    {

      "type": "REMOVE_CONTENT",

      "contentUid": "ANDROID-10:A5:D0:FC:24:9Be",

      "missionName": "train",

      "timestamp": "2016-05-02T14:08:45.371Z",

      "creatorUid": "bob"

    },
```

```
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "e",
      "missionName": "train",
      "timestamp": "2016-05-02T14:07:08.158Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "d",
      "missionName": "train",
      "timestamp": "2016-05-02T14:07:04.637Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "c",
      "missionName": "train",
      "timestamp": "2016-05-02T14:07:00.860Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "b",
      "missionName": "train",
      "timestamp": "2016-05-02T14:06:55.369Z",
      "creatorUid": "bob"
    },
```

```
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "a",
      "missionName": "train",
      "timestamp": "2016-05-02T14:06:50.444Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentUid": "anon1",
      "missionName": "train",
      "timestamp": "2016-05-02T13:50:52.875Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentResource": {
        "hash": "anon1"
      },
      "missionName": "train",
      "timestamp": "2016-05-02T13:50:38.877Z",
      "creatorUid": "bob"
    },
    {
      "type": "REMOVE_CONTENT",
      "contentResource": {
        "hash":
"2cbcbbee4f8c2895b1a079281543343ef4e0b2db4505b6a6056ebec5ea2cf336"
```

```
      },
      "missionName": "train",
      "timestamp": "2016-04-22T15:34:33.164Z",
      "creatorUid": ""
    },
    {
      "type": "ADD_CONTENT",
      "contentResource": {
        "keywords": [],
        "name": "Wide_photo.kmz",
        "submissionTime": "2016-04-22T15:31:02.829Z",
        "submitter": "admin",
        "uid": "9698hhsg",
        "creatorUid": "yyy666",
        "hash":
"78cd540c8e352e6ff416fc1ec48de602c20a6274d1c1201409a5fbb107f23f8f",
        "size": 55981
      },
      "missionName": "train",
      "timestamp": "2016-04-22T15:31:47.800Z",
      "creatorUid": "joe1234"
    },
    {
      "type": "CREATE_MISSION",
      "missionName": "train",
      "timestamp": "2016-04-22T15:26:17.260Z",
      "creatorUid": "bob"
    }
```

```
  ]

}
```

Error Response Code: 404 Not Found

Resource: Mission Keywords

PUT (add keywords 'bunker' and 'minivan' to mission 'cambridge')

```
curl -X PUT
"http://<username>:<password>@localhost:8080/Marti/api/missions/cambri
dge/keywords" -d '["bunker", "minivan"]' -H "Content-Type:
application/json"
```

Required Request Headers:

```
Content-Type: application/json
```

Request Body:

A JSON array of keyword strings (double-quoted.)

```
["bunker", "minivan"]
```

Success Response Code: 200 OK

```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "cambridge",

      "createTime": "2016-01-13T15:46:30.853Z",

      "contents": [],

      "keywords": [

        "bunker",

        "minivan"
```

```
      ]

    }

  ]

}
```

Error Response Code: 404 Not Found (if the mission doesn't exist)

DELETE (clear all keywords for mission 'cambridge')

```
curl -X DELETE
"http://<username>:<password>@localhost:8080/Marti/api/missions/cambri
dge/keywords"
```

Success Response Code: 200 OK

```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "cambridge",

      "createTime": "2016-01-13T15:58:19.887Z",

      "contents": [],

      "keywords": []

    }

  ]

}
```

Error Response Code: 404 Not Found (if the mission doesn't exist)

DELETE (remove keyword 'minivan' from mission 'cambridge')

```
curl -X DELETE
"http://<username>:<password>@localhost:8080/Marti/api/missions/cambri
dge/keywords/minivan"
```

Success Response Code: 200 OK


```
{

  "version": "1.0.0",

  "type": "Mission",

  "data": [

    {

      "name": "cambridge",

      "createTime": "2016-01-13T15:58:19.887Z",

      "contents": [],

      "keywords": []

    }

  ]

}
```


Error Response Code: 404 Not Found (if the mission doesn't exist)

Resource: Mission Logs

Mission Logs (Recce Logs) are composed of free-form text entries which can be associated with one or more missions. Log entries can be created, read, updated, deleted. The complete log for a mission may also be downloaded, consisting of all log entries for the mission.

POST: Create a log entry

This is a POST (and not a PUT) because the log entry unique identifier is created by the server.
Path: api/missions/logs/entries

HTTP Method: POST

Request Headers (required)
Content-Type : application/json;charset=UTF-8

Request Body

content : The text of the log entry
creatorUid : Client UID
entryUid : UID of associated CoT track
missionNames : Missions to associate with this log entry
contentHashes : Hashes of enterprise sync files to associate with this log entry
dtg : User-provided timestamp for this log entry (corresponding to what's described in the text of the log entry, not a server-side timestamp)

```
{

    "content" : "Coffee is delicious.",

    "creatorUid" : "bob",

    "entryUid" : "abc123",

    "missionNames" : ["pittsburgh"],

    "contentHashes" : ["463g7", "6778g6s"],

    "dtg" : "2010-02-10T00:00:00.000Z"

}
```

Success response code: 201 CREATED

Example: Add a log entry to the "Pittsburgh" mission log

```
curl -v -X POST '' -d '{

    "content" : "Coffee is delicious.",

    "creatorUid" : "bob",

    "entryUid" : "abc123",

    "missionNames" : ["pittsburgh"],

    "contentHashes" : ["463g7", "6778g6s"],

    "dtg" : "2010-02-10T00:00:00.000Z"

}'
http://<username>:<password>@localhost:8080/Marti/api/missions/logs/en
tries"
```

Success response body:

The response is a reflection of the incoming request - with the significant addition of the id, which is a pseudo-random 128-bit identifier assigned by the server at entry creation time, and "servertime" - the server-side timestamp at creation time.


```
{
  "version": "1.0.0",
  "type": "com.bbn.marti.sync.model.LogEntry",
  "data": {
    "id": "df9c8b5d-e952-408c-b554-afd6c6a2ba99",
    "content": "Coffee is delicious.",
    "creatorUid": "bob",
    "entryUid": "abc123",
    "missionNames": [
      "pittsburgh"
    ],
    "servertime": "2016-05-03T18:00:30.599Z",
    "dtg": "2010-02-10T00:00:00.0Z",
    "contentHashes": [
      "463g7",
      "6778g6s"
    ]
  }
}
```


GET: Download a mission log

Path: api/missions/pittsburgh/log

HTTP Method: GET

Query Parameters *Optional*

secago : time interval - number of seconds in the the past, until now.
start : start of time interval. Takes precedence over "secago" if specified.
end : end of time interval

No request body.

Success response code: 200 OK

Success response body:


```
{
  "version": "1.0.0",
  "type": "com.bbn.marti.sync.model.LogEntry",
  "data": [
    {
      "id": "df9c8b5d-e952-408c-b554-afd6c6a2ba99",
      "content": "Coffee was delicious.",
      "creatorUid": "bob",
      "entryUid": "abc123",
      "missionNames": [
        "pittsburgh"
      ],
      "servertime": "2016-05-03T19:20:57.11Z",
      "dtg": "2010-02-10T00:00:00.0Z",
      "contentHashes": [
        "463g7",
        "6778g6s"
      ]
    },
    {
      "id": "7026c588-0169-47e7-bd50-2d879482f01b",
      "content": "Snacks are great.",
```

```
      "creatorUid": "bob",
      "entryUid": "abc123",
      "missionNames": [
        "pittsburgh"
      ],
      "servertime": "2016-05-03T19:55:57.478Z",
      "dtg": "2010-02-10T00:00:00.0Z",
      "contentHashes": [
        "463g7",
        "6778g6s"
      ]
    },
    {
      "id": "6de75010-dfa4-467f-b095-f16fd219a7b7",
      "content": "bookshelf",
      "creatorUid": "bob",
      "entryUid": "abc123",
      "missionNames": [
        "pittsburgh"
      ],
      "servertime": "2016-05-03T19:58:54.575Z",
      "dtg": "2017-02-10T00:00:00.0Z",
      "contentHashes": [
        "463g7",
        "6778g6s"
      ]
    }
  ]
```

```
}
```

Example: Download a mission log for mission "Pittsburgh"

```
curl -v
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/log"
```

Example: Download a mission log for mission "Pittsburgh", for the last 40 seconds only

```
curl -v
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/log?secago=40"
```

Example: Download a mission log for mission "Pittsburgh", for the given time interval

```
curl -v
"http://<username>:<password>@localhost:8080/Marti/api/missions/pittsb
urgh/log?start=2016-04-01T00:00:00.000Z&end=2016-04-01T20:04:38.817Z
```

Error Responses: 400 Bad Request : invalid request parameters

DELETE: Delete a log entry

The id of the log entry to be deleted is specified in the request URI.

Path: api/missions/logs/entries/{entry_id}

HTTP Method: DELETE

No request body, no response body

Example: Delete a log entry

```
curl -v -X DELETE
http://<username>:<password>@localhost:8080/Marti/api/missions/logs/en
tries/39ff3721-8ddf-4d1d-a8e3-9ecf6c6badf2
```

PUT: Update a log entry

Path: api/missions/logs/entries

HTTP Method: PUT

Request Headers (required) Content-Type : application/json;charset=UTF-8

Example: Update a log entry in the "Pittsburgh" mission log

```
curl -v -X PUT '' -d '{

    "id" : "39ff3721-8ddf-4d1d-a8e3-9ecf6c6badf2",

    "content" : "Coffee was delicious.",

    "creatorUid" : "jimmy",

    "entryUid" : "abc123",

    "missionNames" : ["pittsburgh"],

    "contentHashes" : ["463g7", "6778g6s"],

    "dtg" : "2010-02-10T00:00:00.000Z"

}'
http://<username>:<password>@localhost:8080/Marti/api/missions/logs/en
tries"
```

Request Body

id : The unique id for the log entry (from the original post, or the mission log) content : The text of the log entry
creatorUid : Client UID
entryUid : CoT UID to associate with this log entry
missionNames : Missions to associate with this log entry
contentHashes : Hashes of enterprise sync files to associate with this log entry
dtg : User-provided timestamp for this log entry (corresponding to what's described in the text of the log entry, not a server-side timestamp)

```
{

    "content" : "Coffee is delicious.",

    "creatorUid" : "bob",

    "entryUid" : "abc123",

    "missionNames" : ["pittsburgh"],

    "contentHashes" : ["463g7", "6778g6s"],

    "dtg" : "2010-02-10T00:00:00.000Z"

}
```

Success response code: 201 CREATED

Success response body:

The response is a reflection of the incoming request - with the significant addition of the id, which is a pseudo-random 128-bit identifier assigned by the server at entry creation time.


```
{
  "version": "1.0.0",
  "type": "com.bbn.marti.sync.model.LogEntry",
  "data": {
    "id": "df9c8b5d-e952-408c-b554-afd6c6a2ba99",
    "content": "Coffee was delicious.",
    "creatorUid": "bob",
    "entryUid": "abc123",
    "missionNames": [
      "pittsburgh"
    ],
    "servertime": "2016-05-03T19:20:57.11Z",
    "dtg": "2010-02-10T00:00:00.0Z",
    "contentHashes": [
      "463g7",
      "6778g6s"
    ]
  }
}
```


GET: Download a combined mission log for all missions

Mainly for QA.

Path: api/missions/all/logs

HTTP Method: GET

No request body.

Success response code: 200 OK

Error Responses: None

Mission Contents Search

Mission contents can be filtered by metadata, such as time interval and bounding box.

example: Filter contents of mission "pittsburgh" by time interval (2016-01-26T21:42:35Z - 2016-01-26T23:00:00Z)

```
curl -v
"http://<username>:<password>@localhost:8080/Marti/api/sync/search?sta
rtTime=2016-01-26T21%3A42%3A35Z&endTime=2016-01-
26T23%3A00%3A00Z&mission=pittsburgh"
```

example: Filter contents of mission "pittsburgh" by geographic bounding box "-73 40, -70 44"

```
curl -v
"http://<username>:<password>@localhost:8080/Marti/api/sync/search?box
=-73%2040%2C%20-70%2044&mission=pittsburgh"
```

Path: api/sync/search

HTTP Request Method: GET

Query Parameters (URL-encoded)

box : geographic bounding box
startTime : time interval start (ISO-8601 Timestamp **with milliseconds**)
endTime : time interval end (ISO-8601 Timestamp **with milliseconds**)
mission : Mission name

Success Response Code: 200 OK


```
{

  "version": "1.0.0",

  "type": "Resource",

  "data": [

    {

      "filename": "",
```

```
      "keywords": [],
      "mimeType": "image\/png",
      "name": "",
      "submissionTime": "2016-01-26T21:59:24.501Z",
      "submitter": "",
      "uid": "681120c332414900acecd5a5cbec970f",
      "hash": "12fg",
      "size": 918779662
    },
    {
      "filename": "",
      "keywords": [],
      "mimeType": "image\/png",
      "name": "",
      "submissionTime": "2016-01-26T21:59:24.501Z",
      "submitter": "",
      "uid": "e99fc9c4e10949058da208e25d842e07",
      "hash": "789n",
      "size": 925329953
    }
  ]
}
```


Files / Enterprise Sync

Enterprise Sync can be used to upload, download and delete files via an HTTP interface.

Related Web UI

Upload File http://localhost:8080/Marti/upload.jsp

List Files http://localhost:8080/Marti/EnterpriseSync.jsp

Upload a File

Example: Upload a PNG image file from the current directory named "map_a.png". Associate it with keywords dog, cat and bird. Specify latitude, longitude and altitude and creatorUid. NB: curl automatically adds the Content-Length header, and sends the "raw" binary file as the request body. Enterprise Sync will also accept uploads via Multi-part MIME, but sending files this way is simpler.

```
curl -v -X POST -H "Content-Type: image/png" --data-binary
"@map_a.png"
'http://<username>:<password>@localhost:8080/Marti/sync/upload?name=ma
p_a.png&keywords=dog%2Ccat%2Cbird&latitude=42.3897510&longitude=-
71.1474660&altitude=10&uid=9698hhsg&creatorUid=anon1'
```

Request Headers (required)

Content-Type : <MIME Type>
Content-Length : <File size (bytes)>

Query Parameters (URL-encoded)

required
name: Filename

optional
uid: UID to associate with the file. If no UID is specified, one will be generated by the server.
latitude: Decimal latitude
longitude: Decimal longitude
altitude: Decimal altitude
keywords: A comma-separated list of search keywords to associate with the file.
creatorUid: The uid of the user uploading the file.

Request Body: The file, in bytes

Success Response Code: 200 OK

Response Body

```
{

  "UID": "9698hhsg",

  "SubmissionDateTime": "2016-01-25T17:30:03.607Z",

  "Keywords": [

    "dog",
```

```
    "cat",

    "bird"

  ],

  "MIMEType": "image\/png",

  "SubmissionUser": "admin",

  "Latitude": "42.3897510",

  "Hash":
"673e62c01ba47a43700a4edb15cef4f6b13f24fb4a7da2fc728368090d3da8f9",

  "Longitude": "-71.1474660",

  "Altitude": "10",

  "Name": "map_a.png"

}
```


Error Responses

400 Bad Request

Download a File

Example: Download the file from the upload example, referencing it by hash

```
curl -v
'http://<username>:<password>@localhost:8080/Marti/sync/content?hash=6
73e62c01ba47a43700a4edb15cef4f6b13f24fb4a7da2fc728368090d3da8f9'
```

Path:
sync/content

HTTP Method: GET

Query Parameters (URL-encoded)

required
hash: The file hash

Success Response Code: 200 OK

Delete a File

Example: Delete the file from the upload example, referencing it by hash

```
curl -v -X POST
'http://<username>:<password>@localhost:8080/Marti/sync/delete?hash=67
3e62c01ba47a43700a4edb15cef4f6b13f24fb4a7da2fc728368090d3da8f9'
```

Path:
sync/delete

HTTP Method: POST

Query Parameters (URL-encoded)

required
hash: The file hash

Success Response Code: 200 OK

Mission Subscriptions and Change Notifications

Subscriptions to missions are tracked (in-memory) by the TAK Server core services, on a "per client UID" basis. For a client to receive updates, a call must be made to explicitly subscribe. Explicitly unsubscribing from updates is also supported. In addition, when a client disconnects, any mission subscriptions for that client will be automatically cancelled, which can be considered an "implicit unsubscribe".

After the subscribe call is processed by the server (which should be nearly instantaneous), the broker will deliver mission change CoT messages.

Subscribe to a Mission

Path: /api/missions/<missionName>/subscription

HTTP Method: PUT

no request body, no response body.

Query Parameters (URL-encoded)

Either uid or topic required

uid : The client UID of the EUD to subscribe

topic : The topic of the WebCOP session to subscribe (See Topic API docuementation)

Success Response Code: 201 Created

Error Response Code: 404 Not Found (If the mission doesn't exist or has been deleted)

Example: Subscribe the client uid 'anon1' to mission 'idaho'.

```
curl -v -X PUT
'http://<username>:<password>@localhost:8080/Marti/api/missions/idaho/
subscription?uid=anon1'
```

Unsubscribe from a Mission

Path: /api/missions/<missionName>/subscription

HTTP Method: DELETE

no request body, no response body.

Query Parameters (URL-encoded)

required

uid : The client uid of the EUD to unsubscribe

uid : The topic uid of the EUD to unsubscribe

Success Response Code: 200 OK

Error Response Code: 404 Not Found (If the mission doesn't exist or has been deleted)

Example: Unsubscribe the client uid 'anon1' from mission 'idaho'.

```
curl -v -X PUT
'http://<username>:<password>@localhost:8080/Marti/api/missions/idaho/
subscription?uid=anon1'
```

Get Mission Subscriptions

Get a list of a subscriptions to missions. Mainly for debugging purposes.

```
curl
"http://<username>:<password>@localhost:8080/Marti/api/missions/all/su
bscriptions"
```

Success Response Code: 200 OK

Change Notifications

Once a client has subscribed to a mission, it will receive CoT notifications of changes to the mission. The "mission change" CoT messages will be send to the client UID for which a subscriptions has been registered.

Example: A change has occurred in the mission "idaho".

```
<?xml version="1.0" encoding="UTF-8"?>

<event how="h-g-i-g-o" type="t-x-m-c" version="2.0" uid="9d262d23-
2760-4df4-8bc6-aef084d2fdff" start="2016-03-16T09:45:35.716Z"
time="2016-03-16T09:45:35.716Z" stale="2016-03-
16T09:45:55.716Z"><point ce="9999999" le="9999999" hae="0" lat="0"
lon="0"/><detail><mission type="CHANGE"
name="idaho"/></detail></event>
```

Change notifications will occur for the following events

- Create Mission
- Delete Mission
- Add Content to Mission (by hash or uid)
- Create, Update or Delete Mission Log Entry. For mission log changes, the notification CoT type will be "t-x-m-c-l"

Data Sync Flow Examples

Create, populate and filter mission

Get missions

If the "Idaho" mission doesn't exist, Create mission "Idaho".

Upload mission files

Add files to mission

Search mission by geographic bounding box filter

Search mission by time range filter

Download in other formats:

- Plain Text

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